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Untangling our checkered past: Investigating the link between local historic district designation and spatial segregation history

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ABSTRACT

The growing requests for equitable and inclusive historic preservation raise questions about how the designation of local historic districts has been practiced over the last 50 years. While several studies have examined neighborhood effects after designating local historic districts, empirical research investigating the association of neighborhood characteristics with local historic districts in the designation process remains limited. To address this, we explored the prevalence of local historic district designations in historically redlined or recently segregated neighborhoods compared to non-redlined or non-segregated neighborhoods in 32 large U.S. cities. We used the Home Owners' Loan Corporation (HOLC) maps as a proxy of historical redlining and measured the Getis-Ord local *Gi** statistic to identify recent segregation patterns. Our analysis using the *Gi** statistic showed significant associations with local historic district designations, whereas the results from the HOLC models were largely inconclusive. We found a higher likelihood of local historic district designations in racially or economically segregated neighborhoods between the 1970s and 1990s, but a potential shift was observed in the 21st century. These findings highlight the need to consider historical redlining contexts and contemporary patterns of segregation in local historic district designations to ensure that preservation policies do not reinforce existing spatial disparities.

1. Introduction

Local historic districts in the U.S. have served as an important neighborhood planning tool beyond the primary purpose of protecting an ensemble of historic resources since the passage of the National Historic Preservation Act in 1966 (Birch & Roby, 1984; Minner, 2016). Their designation acts as a certification of historic significance and a prerequisite for policy interventions to control transformations in the built environment at the neighborhood scale (Hamer, 1998; Schuster, 2002). Compared to the National Register of Historic Districts without any legal obligations, local historic district designation is generally followed by regulatory controls over any exterior changes, new construction, or demolition within the district boundaries. Local historic districts often face criticism due to the stringent and cumbersome legal obligations property owners must comply with (Anderheggen, 2010; Heuer, 2007). On the other hand, strict regulations contribute to increased property values by protecting the unique characteristics of neighborhoods and managing the pace of neighborhood transformation (Been et al., 2016; Coulson & Lahr, 2005; Zahirovic-Herbert & Chatterjee, 2012). The positive impact of local historic districts on neighborhood stabilization and neighborhood character preservation results in continuous designations of local historic districts in many American cities even today.

The growing requests for equitable and inclusive historic preservation, both in the U.S. and worldwide, raise questions about how preservation policies have been practiced in systemically segregated neighborhoods and historically marginalized communities (Arlotta & Avrami, 2020; Cheong, 2021; Hosagrahar, 2018; Minner, 2016; Rastegar et al., 2021). These questions are connected with the equity preservation framework, which requires inclusive processes and equitable outcomes in the relationship between preservation practices and local communities. Several studies examining the outcomes of historic preservation have demonstrated that local historic districts are more likely to affect property values positively (Been et al., 2016; Coulson & Leichenko, 2004; Heintzelman & Altieri, 2013; Noonan, 2007; Noonan & Krupka, 2011; Zahirovic-Herbert & Chatterjee, 2012), and thereby increase the possibility of neighborhood socioeconomic change (Choi, 2025; Kinahan & Ruther, 2020; McCabe & Ellen, 2016). On the other

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hand, the designation processes have been challenged for the underrepresentation of the histories of communities of color and their cultural heritage (Hayden, 1988; T. Lee, 2012). However, there is a notable lack of empirical research investigating how neighborhood socioeconomic characteristics are associated with the creation of local historic districts (Noonan & Krupka, 2010).

To address this research gap, we examine the prevalence of local historic district designations in historically redlined or recently segregated neighborhoods compared to non-redlined or non-segregated neighborhoods in 32 large U.S. cities. This study can contribute to a recent discussion of the equity preservation framework by providing a nuanced understanding of the relationship between historic district designations and residential segregation. This study uses the Home Owners' Loan Corporation (HOLC) maps representing historical redlining practices and the Getis-Ord Local G_i* statistic as a proxy of recent patterns of spatial segregation, suggesting the potential of digitized archival data and a segregation index to assess preservation practices in terms of equity and inclusivity (Hosagrahar, 2018; Landorf, 2011). The findings from our analysis have broad implications for preservation policy and neighborhood planning in segregated neighborhoods. Despite our focus on U.S. cases, this research offers valuable insights for those who work toward an equity preservation approach in urban areas worldwide that face similar challenges of inclusivity and spatial justice. In the U.S., our findings suggest that urban planners and policymakers can integrate equity and inclusive considerations into historic designation procedures, particularly considering historically institutionalized spatial injustice that persists in contemporary urban landscapes. Internationally, this study underscores the need to promote diversity and inclusion in historic designation procedures through fair representation and inclusive participation of underrepresented communities.

2. Designation of local historic districts

Charleston's Old and Historic District, established in 1931, was the first local historic district nationwide to manage transformations in an old residential neighborhood through the use of the zoning ordinance (Weyeneth, 2003). In the mid-20th century, several municipalities adopted the Charleston model that designates local historic districts with requirements of design guidelines and board reviews. The widespread establishment of local historic districts across the country was spurred by the passage of the National Historic Preservation Act of 1966, which established the legal framework for federal preservation activities and encouraged local preservation efforts (Tyler et al., 2018). In addition, federal community development policies including the Model Cities Program of 1966 and the Neighborhood Development Program of 1968 encouraged neighborhood conservation as one of the strategies for community building and welfare (Birch & Roby, 1984; Tomlan, 2015). Hence, the number of locally designated historic districts across the country rapidly increased in a short period from approximately a hundred in 1972 to roughly two thousand in 1986 (Howard, 1988). Today, many local governments continue to create local historic districts to preserve the character and cultural heritages of historic neighborhoods.[1]

The designation process of a local historic district based on a municipal preservation ordinance generally consists of three phases: nomination, review, and designation. First, cities usually allow anyone to nominate a specific area as a local historic district if the area is considered historically significant. The criteria for historic significance vary by city, but most municipalities model the four types of significance for the National Register of Historic Places (event, people, architectural design, and archeology). Second, a review procedure for establishing local historic districts is typically administered by the local preservation board in coordination with preservation planners. The board also holds a public hearing to assess historic significance of the area and hear opinions from a variety of stakeholders. At the end, the board members vote on whether to designate the area as a local historic district. In addition, all property owners within the nominated area are notified for designation approval. Lastly, the designation of a local historic district is included as a part of a zoning ordinance after the approval of the city council and mayor.

After the establishment of a local historic district, regulations and incentives come into effect. Generally, the regulatory power to review and approve all proposals for exterior changes in the district is granted to municipal preservation boards or local district commissions. The exterior changes include restorations, alterations, additions, demolitions, or new construction. Most of their decisions are generally based on the *Secretary of Interior's Standards for Rehabilitation.* Local design guidelines or district ordinances also provide detailed standards applicable to the unique context of a specific historic district. Meanwhile, many municipalities adopted various incentive programs to encourage preservation activities in local historic districts. Financial incentives offered for the maintenance and rehabilitation of historic properties include grants or subsidies, income tax credits, property tax reductions, and fee exemptions. Planning incentives also exist such as zoning/building code relief, incentive zoning for extra floor area or height, and transfer of development rights.

3. District designation and residential segregation

There is a widespread call for more equitable and inclusive historic preservation, both in the U.S. and worldwide, as preservation processes and outcomes have been criticized for reinforcing existing spatial and socioeconomic inequalities. In particular, prioritizing material integrity and tangible heritage often makes preservation policies and practices appear elitist and expert-driven, excluding underrepresented groups and their narratives in decision-making processes. Such exclusion can raise significant concerns regarding the fair distribution of preservation outcomes, which tends to favor the benefits of decision-makers rather than the well-being of communities in and around cultural heritage. In response to these critiques, there have been international efforts, along with the United Nations 2030 Agenda for Sustainable Development, to integrate equity and inclusiveness principles into the management of World Heritage (Labadi et al., 2021; UNESCO, 2015). In order to achieve the equity goals as one of the 3Es of sustainable development (the economy, the environment, and equity; Campbell, 1996), the application of the social justice framework in heritage management has also been discussed for promoting inclusive procedures that prioritize community needs and enhance community empowerment (Baird, 2014; Rastegar et al., 2021). Likewise, in the American context, with the close connection between historic preservation and urban planning, several scholars have explored agendas for equitable and inclusive preservation practices based on the ideas of sustainable development (Avrami, 2016) and socially just planning (Cheong, 2021; Minner, 2016).

According to a substantial body of previous literature (Arlotta & Avrami, 2020; Baird, 2014; Cheong, 2021; Hosagrahar, 2018; Minner, 2016; Rastegar et al., 2021), the equity preservation framework highlights the importance of inclusive processes and equitable outcomes to ensure socially just preservation practices and embrace cultural diversity in local communities with and around cultural heritage and historic resources (Fig. 1). Inclusive procedures require the participation of diverse stakeholders throughout the entire preservation process, including identifying, designating, managing, and utilizing historic resources. In particular, historically marginalized and disadvantaged communities receive attention regarding the representation of their histories and narratives and their active participation in decision-making processes and preservation practices. In addition, equitable outcomes consider the impacts of historic preservation on local communities and surrounding areas, with particular attention to how the

[1] Even though a significant increase in local historic districts is expected since the estimation by Howard (1988), there has not been any updated data on the exact number of local historic districts across the country.

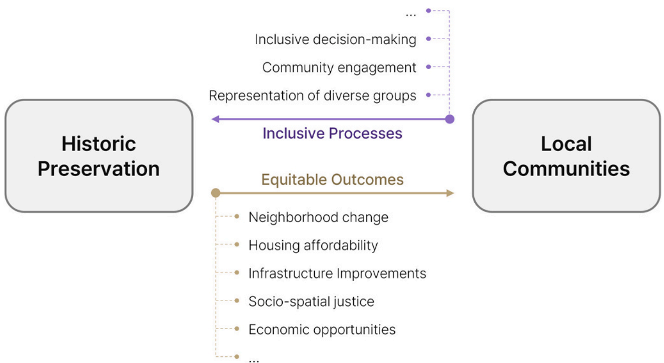

Fig. 1. The equity preservation framework in relation to local communities.

benefits and costs of preservation practices are distributed. These questions about fair distributions lead to the broader challenge of how to balance historic preservation with various contemporary urban concerns such as neighborhood change, housing affordability, infrastructure improvements, and socio-spatial justice. This focus on inclusive processes and equitable outcomes within the equity preservation framework provides an important foundation for exploring the relationship between the designation of historic districts and the spatial patterns of residential segregation.

The persistent pattern of residential segregation in American cities has been created and reinforced by a long history of institutionalized discriminatory housing policies and planning practices (Rothstein, 2017). The enduring negative impact of segregation on urban environments and socioeconomic characteristics of neighborhoods is confirmed by the persistent disparities in resources between segregated and unsegregated neighborhoods (Aaronson et al., 2021; Jacoby et al., 2018; Jung et al., 2024; Rutan & Glass, 2018; White et al., 2021; Wilson, 2020). However, a significant research gap remains on whether preservation policies and practices have contributed to exacerbating or challenging the persisting spatial patterns of exclusion embedded in urban landscapes (Arlotta & Avrami, 2020). Despite limited research examining the relationship between historic district designation and residential segregation, the equity preservation framework provides a lens to explore several themes from existing preservation literature that suggests potential connections. First, segregated neighborhoods might be underrepresented in historic designation procedures, highlighting a potential lack of inclusive processes. Second, historic districts may exacerbate segregation through gentrification and displacement, further challenging equitable outcomes. Third, their restrictive and exclusive character may directly reinforce residential segregation.

The designation of historic resources as a primary preservation tool has faced ongoing criticism for its failure to represent diverse population groups and their cultural heritage equitably (Hayden, 1988; T. Lee, 2012). Critics argued that the history of underrepresented populations makes up a very small portion of all national and local lists. The National Park Service (2014) estimated that less than 8 % of the National Registers and National Historic Landmarks are relevant to non-white populations.[2] A group of case studies that explored the dynamics in the designation process found several structural challenges that led to the underrepresentation of segregated communities. First, the deep-rooted prejudice against under-resourced neighborhoods labeled as “blighted” and “slums” is pervasive among experts, city officials, and members of landmark committees (Michael, 2005; Saito, 2009). This prejudice leads to undervaluing the historic significance of cultural heritage in these neighborhoods and undermining the need to preserve these areas. In addition, the designation criteria emphasizing physical integrity pose a challenge for properly evaluating the historic resources within marginalized communities because much of their significance lies in intangible features (Buckley & Graves, 2016; Hamer, 1998; Kaufman, 2013). Most of the local designation standards influenced by the National Register criteria prioritize the material integrity and architectural merits of a historic property. Finally, underrepresented communities are left out of the preservation decision-making process (Zhang, 2011). Local political structure often limits the influence and involvement of communities of color in designation decisions as they are generally disempowered in discussions concerning land use and urban development.

The neighborhood effects of historic district designation in segregated neighborhoods are another concern from the equity preservation framework in terms of the distribution of benefits and costs. With the continuous focus on the economic benefits of historic preservation among advocates, historic districts are considered a primary preservation tool for community development (Listokin et al., 1998; Mason, 2005; Ryberg-Webster & Kinahan, 2014; Rypkema et al., 2013). In particular, prior studies indicate the positive impacts of local historic districts on property values due to the strict regulations on external changes that contribute to minimizing uncertainty around future neighborhood transformations (Been et al., 2016; Coulson & Lahr, 2005; Zahirovic-Herbert & Chatterjee, 2012). However, the increase in property values, particularly larger in low-priced properties, could bring about the potential risk of gentrification and displacement for people with lower incomes because they are more vulnerable to rising property values and rents (Zahirovic-Herbert & Chatterjee, 2012). The threat of gentrification and displacement in segregated neighborhoods is supported by recent studies on neighborhood change in local historic districts, which found significant changes in the socioeconomic status and racial composition within the districts after designation (Choi, 2025; Kinahan & Ruther, 2020; McCabe & Ellen, 2016).

While the previous discussion focused on the underrepresentation in the designation process and the negative neighborhood effects of historic district designation within historically under-resourced neighborhoods, the designation of local historic districts could exacerbate residential segregation by reinforcing the neighborhood characteristics of separateness and exclusiveness as historically planned features (Hamer, 1998). Some residents of urban neighborhoods expect that local historic districts function as exclusionary zoning to restrict new housing development and displace people with lower incomes (Fein, 1985). Stringent regulations subject to district designations provide controls over neighborhood transformations and impose additional costs on residents. An analysis of local historic districts in New York City reveals a decrease in new construction activities within district boundaries after designation (Been et al., 2016). Although this evidence does not indicate its influence on the overall housing supply in the city, district regulations with other planning restrictions on development may weaken affordability (Glaeser, 2011). Additionally, a recent study on housing sales within local historic districts in Denver identified a higher likelihood of a homebuyer being white and of transactions between white sellers and white buyers, which implies increasing racial homogeneity of neighborhoods (Bologna Pavlik & Zhou, 2023).

While previous literature has suggested a possibility of underrepresentation in district designation processes and the potential impacts of historic districts on residential segregation, a large gap exists in empirical research examining how the designation of historic districts is associated with the spatial patterns of residential segregation. Moreover, there is a lack of understanding regarding which socioeconomic and housing factors influence historic designation decisions (Maskey et al., 2009; Noonan & Krupka, 2010), despite the importance of historic designation as a fundamental preservation tool. Furthermore, developing robust and systemic assessments and metrics to evaluate the equity and inclusivity of preservation policies and practices is imperative globally (Hosagrahar, 2018; Landorf, 2011). Hence, our study fills this research gap by using digitized archival data and spatial statistics to

[2] There is a doubt about this estimation “due to lack of data and the limitations of the current protocol for tracking ethnic associations” (Gao & Dublin, 2020).

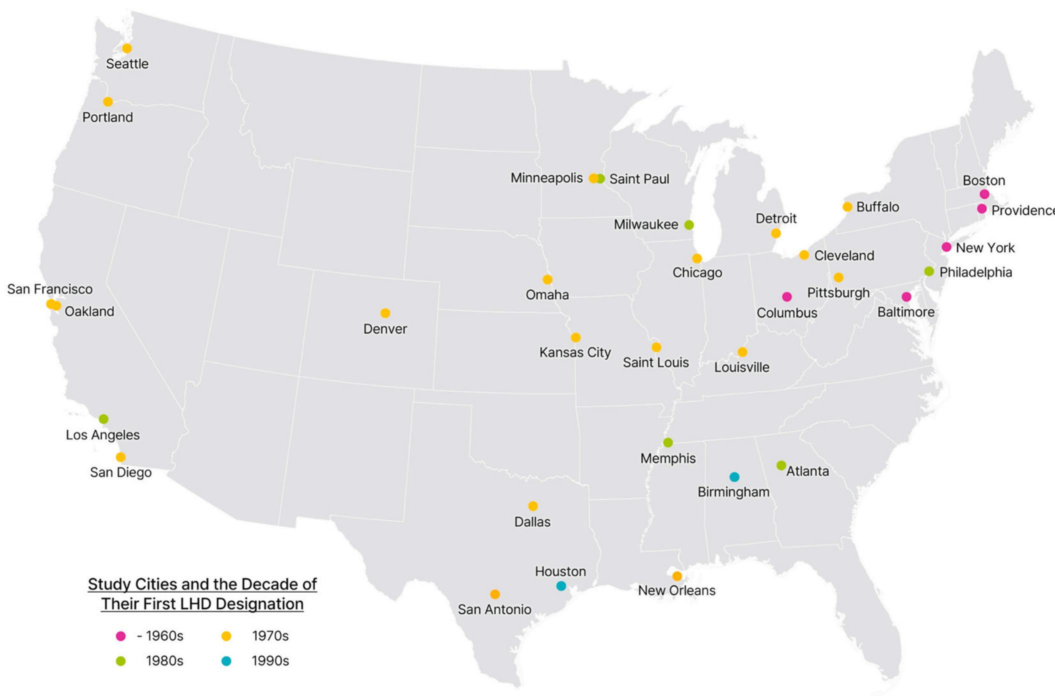

Fig. 2. A map of the 32 Study Cities and the decade of their first local historic district (LHD) designation.

analyze whether segregated neighborhoods are more or less likely to be designated as local historic districts. The findings from this analysis can contribute to the ongoing discourse on the equity preservation framework by exploring the relationship between historic designation and residential segregation through the lens of inclusive processes. Moreover, these findings can provide valuable insights for advancing more socially just preservation planning and informing future designation decisions by evaluating how historic districts were previously designated.

4. Data and methods

4.1. Study areas

We selected 32 large U.S. cities to examine the relationship between locally designated historic districts and historical and recent patterns of spatial segregation (Fig. 2). These cities had a population of over 200,000 in 1940 when the HOLC finished creating neighborhood evaluation maps.[3] This population threshold enables us to select the cities with sufficient numbers of old neighborhoods in each of the four HOLC grades to investigate the association with historical redlining. Spatial and additional data that informs local historic districts is also available for these cities. Based on this data, we analyzed 755 local historic districts designated by the end of 2019, excluding 52 districts in non-residential areas such as streets, parks, cemeteries, campuses, and military bases.

4.2. Measurements on spatial segregation

We used the digitized HOLC residential security maps from the Digital Scholarship Lab at the University of Richmond (Nelson et al., 2018) to investigate the association of local historic district designations with historical redlining. The federal government established the HOLC in the New Deal era to mitigate the increasing rate of housing foreclosures by providing financial relief for homeowners (Hillier, 2003). After finishing their lending program in 1935, HOLC staff members working with local professionals began evaluating the mortgage security of neighborhoods in over 200 cities to create color-coded maps with four grades — grade A for “Best (green),” grade B for “Still Desirable (blue),” grade C for “Definitely Declining (yellow),” and grade D for “Hazardous (red).”

The HOLC neighborhood evaluation maps were considered to institutionalize discriminatory appraisal methods and contribute to inequitable lending decisions (Jackson, 1985). The language used in the rating files created in the evaluations shows that racial and class prejudice was pervasive among the local experts (Crossney & Bartelt, 2005b). It is important to note that recent research reveals lenders did not make loans to certain neighborhoods based on their demographics or housing characteristics before the existence of HOLC surveys, and the HOLC still granted many loans in neighborhoods with poor grades (Crossney & Bartelt, 2005a; Hillier, 2003). Although African Americans were able to access the HOLC refinancing program, the HOLC loans are perceived as reinforcing residential segregation and bringing benefits to white creditors (Michney & Winling, 2020). In addition, the evaluation of most black neighborhoods as hazardous (grade D) demonstrates that the HOLC maps reflect the discriminatory ideas prevalent in a vast network of institutions between academia, private industry, and government in the 1920s and 1930s (Markley, 2023; Michney, 2022; Winling & Michney, 2021). Therefore, the HOLC maps are the primary representation of historic redlining and the most accessible one that covers a large number of cities across the country (Hillier, 2003; Jackson, 1985).

We used the local Getis-Ord *Gi** statistic as a proxy of recent patterns of residential segregation at the time of local historic district designations, which measures the degree of spatial association around a specific area *i* (Getis & Ord, 1992). It assesses whether high or low values of a feature are clustered within a specified distance from area *i* by comparing the sum of values in the neighboring areas within that distance to the overall distribution of values across the entire study area.

[3] Although 40 cities had a population of over 200,000 in 1940, data was unavailable for six cities: Akron, OH, Dayton, OH, Indianapolis, IN, Rochester, NY, Syracuse, NY, and Toledo, OH. Also, two cities (Jersey City, NJ and Newark, NJ) do not have any Grade A neighborhood.

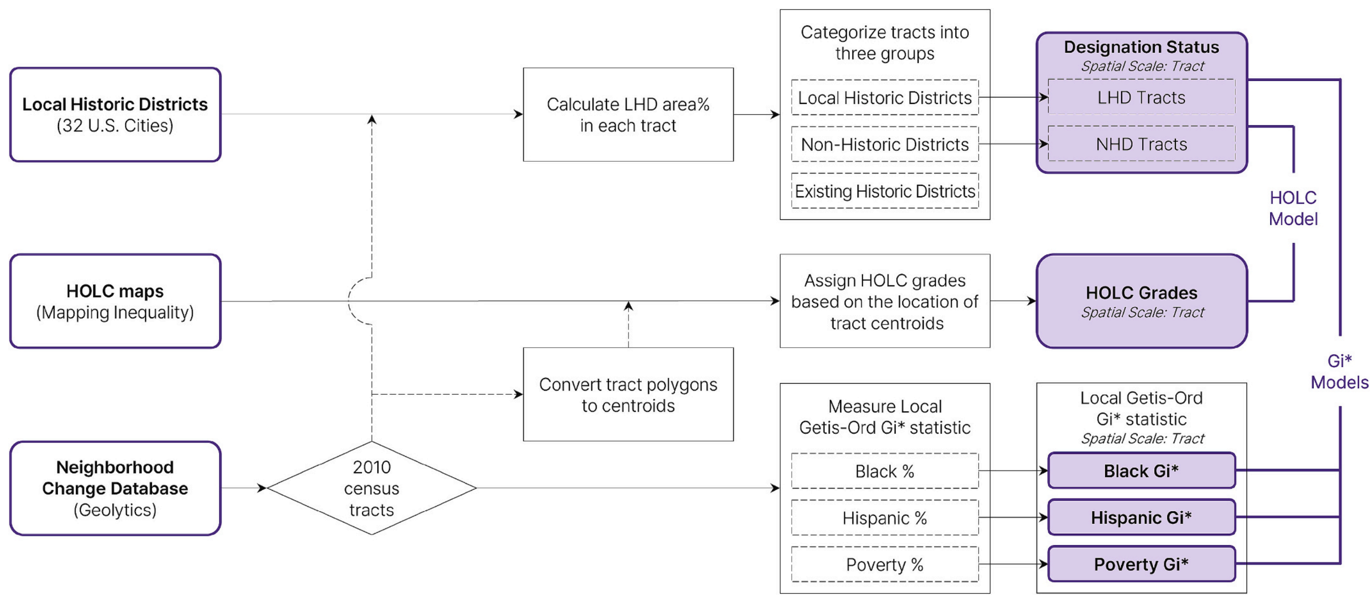

Fig. 3. Research Protocol for processing spatial datasets of each city in each decade for this study.

The local Getis-Ord *Gi** statistic is given as:

$$G_i^* = \frac{\sum_{j=1}^{n} w_{i,j}x_j - \overline{x}\sum_{j=1}^{n} w_{i,j}}{S\sqrt{\frac{n\sum_{j=1}^{n} w_{i,j}^2 - \left(\sum_{j=1}^{n} w_{i,j}\right)^2}{n-1}}}$$

where x_j is represents the value of the feature at area j; $w_{i,j}$ is the spatial weight between areas i and j, indicating the spatial relationship; $\overline{x}$ is the mean of the feature x across all areas; S is the standard deviation of x; n is the total number of areas within the study area.

As a spatially weighted z-score, the local Getis-Ord *Gi** statistic using the racial or socioeconomic composition of each neighborhood accounts for the extent of racial or economic segregation of a specific neighborhood compared to the other neighborhoods within a city or a metropolitan area (Barber et al., 2018; Poulsen et al., 2011). A higher and positive *Gi** statistic suggests that a neighborhood is more racially or economically segregated as it represents a larger cluster of similar racial or economic population groups. Conversely, a lower and negative *Gi** statistic indicates a less segregated neighborhood, indicating a lower concentration of such population groups. This local *Gi** statistic that captures local variation is more pertinent for our research, in comparison with the global segregation indices that only provide a single-index number over a city or a metropolitan area (Massey & Denton, 1988).

4.3. Statistical analysis

We used binary logistic regression to analyze the relationship between local historic district designations and the spatial patterns of historical redlining and recent segregation (Fig. 3). We estimated separate models for each decade between the 1970s and the 2010s to account for different neighborhood characteristics at the times of designation. In particular, some of the cities were excluded from the models for the 1970s and 1980s because their first local historic districts had not been designated by that decade.[4] In order to measure the *Gi** statistic and control for neighborhood socioeconomic and housing characteristics relevant to historic designations in the statistical models, we utilized the Neighborhood Change Database (NCDB). The NCDB normalized the decennial census from 1970 to 2010 and the 2010 American Community Survey into the 2010 census tracts, which were employed as neighborhood proxies for the unit of analysis in this study. Additionally, census tracts with fewer than one hundred residents or less than ten households were eliminated from our samples to minimize the impacts of outliers on our results.

The dependent variable of our model is whether or not a neighborhood is designated as a local historic district. In order to adjust the boundary discrepancies between census tracts and local historic districts, we calculated the proportion of local historic district area in each census tract to categorize census tracts into three different groups: local historic district (LHD), existing historic district (EHD), and non-historic district (NHD) tracts (Table 1). LHD tracts refer to any tract in which more than 5 % of its area was designated as local historic district in each decade. Although setting the larger ratio as the threshold of LHD tracts represents the neighborhood features of local historic districts more accurately, the 5 % threshold enables us to include small historic districts at the neighborhood center in the group of LHD tracts.[5] EHD tracts refer to any tract that was labeled as a LHD tract in the previous decades

Table 1
The number of census tracts by the categories.

Decades (no. of cities/districts)	1970s (23/114)	1980s (30/163)	1990s (32/134)	2000s (32/184)	2010s (32/134)
Local historic district (LHD) tracts	187	189	165	211	220
Non-historic district (NHD) tracts	6144	8022	8429	8283	8059
Existing historic district (EHD) tracts	54	228	398	526	674

[4] Six cities designated their first local historic districts in the 1980s: Atlanta, GA, Los Angeles, CA, Memphis, TN, Milwaukee, WI, Philadelphia, PA, and Saint Paul, MN. In addition, Birmingham, AL and Houston, TX designated their first local historic districts in the 1990s.

[5] A total of 1312 census tracts contain local historic districts within their boundaries. In these tracts, the average ratio of the local historic district area to the tract area is 26.9 %, while the median ratio is 14.6 %. Based on this result, we tested three different thresholds (15 %, 10 %, and 5 %) to categorize LHD tracts. With thresholds set at 15 %, 10 %, and 5 %, the number of LHD tracts is 648 (49.4 %), 753 (57.4 %), and 905 (69.0 %), respectively.

Table 2
Neighborhood characteristics of local historic district (LHD) and non-historic district (NHD) tracts (standard deviation in parentheses).

Census Year	1970		1980		1990		2000		2010	
Tract Categories (No. of tracts)	LHD (187)	NHD (6144)	LHD (189)	NHD (8022)	LHD (165)	NHD (8429)	LHD (211)	NHD (8283)	LHD (220)	NHD (8059)
% Old Buildings*	76.7	53.9	73.6	55.8	74.5	63.0	81.0	73.5	89.2	80.8
	(19.3)	(31.8)	(19.9)	(31.2)	(20.9)	(29.5)	(13.8)	(25.7)	(10.8)	(21.3)
% Age 65 and Older	14.8	11.6	15.5	12.4	13.2	12.4	10.0	11.3	11.0	11.2
	(6.8)	(6.2)	(8.4)	(7.0)	(6.9)	(6.9)	(5.3)	(6.3)	(5.0)	(5.8)
% College or Higher Degree	12.6	10.1	22.2	15.8	33.3	20.4	33.8	24.0	42.9	28.3
	(12.7)	(10.3)	(17.7)	(13.9)	(22.7)	(16.9)	(23.2)	(19.5)	(25.3)	(21.5)
% Original Residents**	14.0	14.6	13.6	16.4	14.8	19.4	14.0	20.9	10.3	9.5
	(7.4)	(9.7)	(8.8)	(11.0)	(10.0)	(12.2)	(8.7)	(11.9)	(5.2)	(6.1)
Population Density (1000/km^2)	11.9	9.6	12.8	8.8	10.6	7.3	8.0	7.7	13.3	7.7
	(12.2)	(11.3)	(11.8)	(10.3)	(14.6)	(9.0)	(8.2)	(9.7)	(14.5)	(9.9)
% Vacant Units	9.3	4.7	10.2	6.4	12.7	8.1	9.0	7.1	12.6	10.5
	(5.6)	(4.6)	(6.1)	(5.6)	(7.7)	(6.4)	(7.6)	(5.9)	(9.1)	(7.9)
% Renter-Occupied Units	72.0	50.3	74.6	55.1	66.8	55.1	69.2	54.3	63.1	55.4
	(24.6)	(28.4)	(19.8)	(26.9)	(19.1)	(25.4)	(18.9)	(25.1)	(18.8)	(23.7)
Distance to City-Center (km)	4.1	10.1	4.5	10.9	4.7	11.2	5.7	11.4	6.6	11.5
	(2.8)	(5.8)	(3.5)	(7.1)	(3.7)	(7.2)	(4.8)	(7.3)	(4.1)	(7.3)

* Old buildings refer to units constructed earlier than 30 years ago.
** Original residents refer to households that moved into their current unit earlier than 20 years ago.

and has not had a new local historic district designation in that specific decade. In our statistical model for each decade, this group of the tracts is excluded from the samples because it is mostly unavailable to designate any new historic districts in the EHD tracts. NHD tracts are the remaining census tracts after categorizing the first two tract groups that do not contain local historic districts.

We separately used two independent variables of the HOLC grades and the local Getis-Ord *Gi** statistic to examine the relationship of local historic district designations to historical redlining and recent segregation patterns. In the analysis using the HOLC grades, each census tract is assigned to one of the HOLC grades by the location of its centroid. Census tracts coded as the HOLC grade D category serve as the reference group to examine whether redlined neighborhoods have more or less local historic districts than non-redlined ones. Although uncoded neighborhoods mostly undeveloped in the early 20th century were excluded, about 75 % of local historic districts are included within the neighborhoods evaluated by the HOLC.

In addition, we measured racial and economic segregation for each decade from the 1970s to the 2010s using the local Getis-Ord *Gi** statistic. Three different neighborhood variables from the NCDB are separately used to measure the *Gi** statistic: the proportion of Black residents, the proportion of Hispanic populations, and the poverty rate. Neighboring census tracts that share edges or corners were weighted, and the jurisdiction of each city was used as the larger areal unit. The relationship of *Gi** statistic based on each neighborhood variable with local historic district designations is individually tested in our statistical models.

In accordance with previous studies (Maskey et al., 2009; Noonan & Krupka, 2010), we controlled for a set of neighborhood variables associated with historic designations. Model 1 is adjusted for historic consideration using the ratio of buildings older than 30 years.[6] Model 2 is further adjusted for the demographic, socioeconomic, and housing characteristics of neighborhoods. Neighborhood demographic and socioeconomic variables include the proportion of persons older than 65 years of age, the proportion of persons with a bachelor's degree or higher educational attainment, the proportion of original residents who have lived in their current unit for more than 20 years, and population density. Additional control variables for neighborhood housing features are the proportion of vacant housing units, the proportion of renter-occupied housing units, and the distance to the city center.

[6] This is the longest period available from the 1970 decennial census data.

5. Results

5.1. Sample characteristics

Table 2 summarizes neighborhood characteristics used in the models to demonstrate differences between LHD and NHD tracts. Overall, the proportion of housing units built over 30 years ago has gradually increased in both LHD and NHD tracts. LHD tracts have higher rates of older buildings than NHD tracts, but the difference has decreased from 21.8 % in 1970 to 8.4 % in 2010. In addition, the share of college-educated residents has increased in both tract types but is larger in LHD tracts. Thus, the education gap had grown from 2.5 % in 1970 to 14.6 % in 2010. On average, the population density in LHD tracts is higher than in that in NHD tracts. Regarding housing characteristics, LHD tracts have more vacant and renter-occupied units than NHD tracts. Finally, LHD tracts are generally located near the city center. The average distance to the city center from LHD tracts is 4–6 km, compared to 10–11 km from NHD tracts.

5.2. The designation of local historic districts by HOLC grades

Fig. 4 shows the proportion of local historic districts by HOLC grades over the five decades, indicating the share of LHD tracts relative to the sum of LHD and NHD tracts within each HOLC grade per decade. Grade A neighborhoods have the highest ratio of locally designated historic districts across all five decades, even though they only comprise approximately 4 % of the total census tracts. In contrast, the ratio of local historic districts in grade C neighborhoods is the lowest despite having a large number of local historic districts. While the share of LHD tracts in grade D tracts is higher than in grade B tracts in the 1970s and 1980s, this trend reverses in the 1990s and 2010s.

Table 3 presents the results from the binominal logistic regressions about the relationship between the creation of local historic districts and the HOLC grades. In the HOLC models, insignificant differences mostly exist between redlined and non-redlined neighborhoods in the designation of local historic districts. In the unadjusted models and the models accounting for the ratio of the older buildings (see Model 1 in Table 3), grade C neighborhoods were likely to have fewer new local historic districts than grade D neighborhoods from the 1970s to the 1990s. Upon further adjustments for neighborhood features (see Model 2 in Table 3), grade B neighborhoods are associated with the likelihood of local historic district designations approximately two times greater than grade D neighborhoods in the 1970s, 1990s, and 2010s.

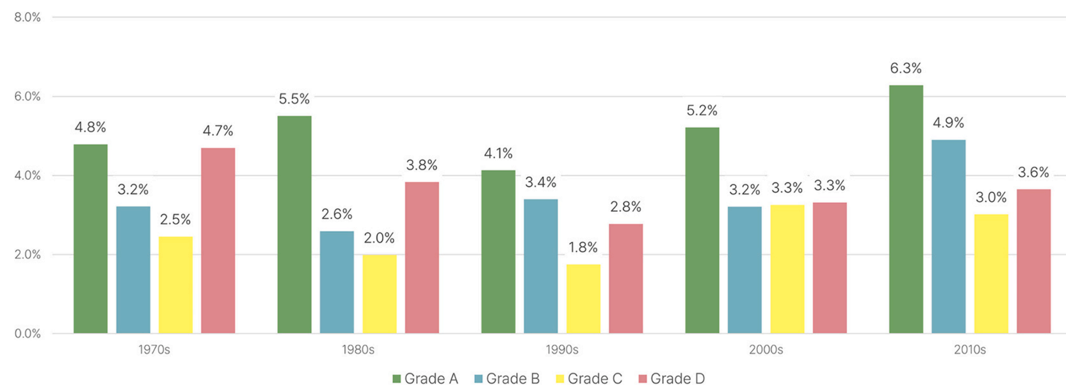

Fig. 4. The proportion of local historic district designations by the HOLC grades.

Table 3
Logit analysis results for the relationship between the designation of local historic districts and the HOLC grades (bold denotes statistical significance, p < 0.05).

	Unadjusted		Model 1		Model 2	
	OR* (95 % CI)	*p-value*	OR (95 % CI)	*p-value*	OR (95 % CI)	*p-value*
1970s (n = 4426)						
Grade A	1.02 (0.48, 2.17)	0.958	1.62 (0.75, 3.51)	0.222	2.51 (0.93, 6.78)	0.069
Grade B	0.67 (0.43, 1.06)	0.090	0.76 (0.48, 1.21)	0.252	**2.09 (1.15, 3.79)**	**0.015**
Grade C	**0.51 (0.35, 0.74)**	**0.000**	**0.53 (0.36, 0.77)**	**0.001**	1.21 (0.78, 1.88)	0.401
Grade D	(ref)		(ref)		(ref)	
1980s (n = 5448)						
Grade A	1.46 (0.78, 2.75)	0.238	1.63 (0.86, 3.07)	0.134	2.19 (0.93, 5.17)	0.074
Grade B	0.67 (0.42, 1.05)	0.082	0.65 (0.41, 1.03)	0.064	1.23 (0.69, 2.21)	0.486
Grade C	**0.51 (0.35, 0.74)**	**0.000**	**0.50 (0.34, 0.73)**	**0.000**	0.84 (0.55, 1.30)	0.438
Grade D	(ref)		(ref)		(ref)	
1990s (n = 5396)						
Grade A	1.51 (0.73, 3.12)	0.269	1.43 (0.69, 2.97)	0.337	1.3 (0.54, 3.16)	0.559
Grade B	1.23 (0.79, 1.92)	0.359	1.15 (0.73, 1.81)	0.557	**1.99 (1.15, 3.46)**	**0.014**
Grade C	**0.62 (0.41, 0.95)**	**0.027**	**0.60 (0.39, 0.91)**	**0.017**	1.10 (0.68, 1.78)	0.691
Grade D	(ref)		(ref)		(ref)	
2000s (n = 5292)						
Grade A	1.60 (0.83, 3.11)	0.164	1.65 (0.85, 3.22)	0.140	2.08 (0.95, 4.52)	0.066
Grade B	0.97 (0.62, 1.51)	0.880	1.01 (0.64, 1.60)	0.949	1.49 (0.89, 2.50)	0.133
Grade C	0.98 (0.69, 1.39)	0.915	1.01 (0.71, 1.45)	0.940	1.21 (0.81, 1.79)	0.352
Grade D	(ref)		(ref)		(ref)	
2010s (n = 5157)						
Grade A	1.77 (0.95, 3.29)	0.071	1.60 (0.86, 2.98)	0.141	1.81 (0.86, 3.81)	0.119
Grade B	1.36 (0.92, 2.01)	0.121	1.17 (0.78, 1.75)	0.439	**1.98 (1.24, 3.18)**	**0.004**
Grade C	0.82 (0.58, 1.17)	0.272	0.74 (0.52, 1.06)	0.097	1.35 (0.91, 2.02)	0.141
Grade D	(ref)		(ref)		(ref)	

* OR, odd ratio; CI, confidence interval; ref., reference category.

5.3. The designation of local historic districts by residential segregation

Fig. 5 shows the proportion of local historic districts by the local Getis-Ord *Gi** statistic measured by the share of Black and Hispanic residents and the poverty ratio. We created segregation categories using a 95 % confidence level of z-score: tracts with the *Gi** statistic above 1.96 are classified as high-segregation neighborhoods, those with the *Gi** statistic between −1.96 and 1.96 as medium-segregation neighborhoods, and those with the *Gi** statistic below −1.96 as low-segregation neighborhoods. Thus, each ratio indicates the proportion of LHD tracts relative to the sum of LHD and NHD tracts within each *Gi** statistic category per decade. Overall, local historic districts were more likely to be designated in racially and economically segregated neighborhoods. Particularly in the 1970s, the proportions of new district designations in high-segregation neighborhoods were around 4–7 %, compared to 0–1 % in low-segregation neighborhoods. However, the proportions of newly designated local historic districts in high-segregation neighborhoods gradually decreased, while there were increases in new locally designated historic districts in low-segregation neighborhoods. Consequently, local historic districts were more established in low-segregation neighborhoods than in segregated neighborhoods around the 2000s and 2010s.

Table 4 presents the results from the binary logit models that examine the relationship between the designation of local historic districts and the recent patterns of spatial segregation measured by the local Getis-Ord *Gi** statistic. Between the 1970s and the 1990s, controlling historic considerations and neighborhood characteristics (see Model 2 in Table 4) reveals a positive correlation between local historic district designation and the *Gi** statistic, suggesting a higher likelihood of locally designated historic districts in racially and economically segregated neighborhoods. In the 2000s and 2010s, this positive relationship persisted only between local historic district designation and

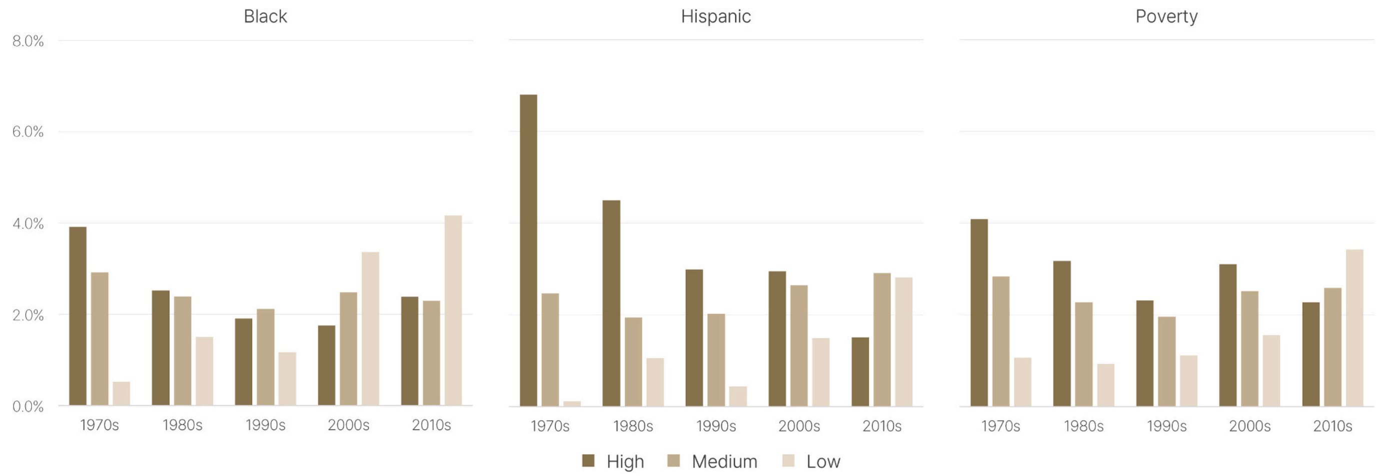

Fig. 5. The proportion of local historic district designations by categories of residential segregation measured by using the Getis-Ord *Gi** statistics. Categories of segregation were defined by a 95 % confidence level of z-score (high: $Gi^* \geq 1.96$, medium: $-1.96 < Gi^* < 1.96$, low: $Gi^* \leq -1.96$).

Table 4
Logit analysis results for the relationship between local district designations and residential segregation (bold denotes statistical significance, $p < 0.05$).

	Unadjusted		Model 1		Model 2	
	OR* (95 % CI)	*p-value*	OR (95 % CI)	*p-value*	OR (95 % CI)	*p-value*
1970s (n = 6331)						
Black	1.13 (1.07, 1.19)	0.077	**1.11 (1.05, 1.17)**	**0.001**	**1.12 (1.04, 1.20)**	**0.003**
Hispanic	**1.13 (1.06, 1.20)**	**0.017**	**1.06 (1.00, 1.13)**	**0.039**	**1.02 (0.95, 1.10)**	**0.041**
Poverty	**1.35 (1.28, 1.42)**	**0.000**	**1.28 (1.21, 1.35)**	**0.000**	**1.14 (1.04, 1.25)**	**0.009**
1980s (n = 8211)						
Black	1.07 (1.02, 1.14)	0.953	**1.05 (0.99, 1.11)**	**0.003**	**1.13 (1.06, 1.22)**	**0.007**
Hispanic	1.12 (1.06, 1.19)	0.125	**1.07 (1.01, 1.14)**	**0.013**	**1.08 (1.00, 1.16)**	**0.015**
Poverty	**1.29 (1.22, 1.36)**	**0.000**	**1.24 (1.17, 1.31)**	**0.000**	**1.18 (1.08, 1.30)**	**0.001**
1990s (n = 8594)						
Black	1.01 (0.95, 1.08)	0.794	1.00 (0.94, 1.07)	0.953	**1.13 (1.04, 1.23)**	**0.003**
Hispanic	**1.07 (1.00, 1.14)**	**0.038**	1.05 (0.99, 1.12)	0.125	**1.11 (1.02, 1.20)**	**0.013**
Poverty	**1.18 (1.11, 1.26)**	**0.000**	**1.18 (1.10, 1.26)**	**0.000**	**1.22 (1.10, 1.36)**	**0.000**
2000s (n = 8494)						
Black	**0.90 (0.84, 0.96)**	**0.002**	**0.90 (0.84, 0.96)**	**0.001**	0.97 (0.89, 1.04)	0.379
Hispanic	**1.08 (1.02, 1.14)**	**0.012**	**1.06 (1.00, 1.13)**	**0.035**	**1.09 (1.02, 1.17)**	**0.016**
Poverty	**1.12 (1.05, 1.19)**	**0.000**	**1.11 (1.05, 1.18)**	**0.001**	1.01 (0.92, 1.12)	0.774
2010s (n = 8279)						
Black	**0.94 (0.88, 1.00)**	**0.041**	**0.93 (0.88, 0.99)**	**0.024**	1.07 (1.00, 1.15)	0.052
Hispanic	0.96 (0.90, 1.02)	0.192	0.95 (0.89, 1.01)	0.085	**1.08 (1.01, 1.17)**	**0.028**
Poverty	0.94 (0.88, 1.01)	0.104	0.94 (0.87, 1.01)	0.082	0.91 (0.82, 1.01)	0.090

* OR, odd ratio; CI, confidence interval; ref., reference category.

the *Gi** statistic measured by the proportion of Hispanic residents. On the other hand, inconclusive results are observed in the relationship between creating local historic districts and the *Gi** Statistic using the proportion of Black populations or the poverty ratio during the same period. However, neighborhoods with a high concentration of Black populations are less likely to have local historic districts in both the unadjusted model and the model accounting for historic considerations (see Model 1 in Table 4), suggesting a potential shift in trends for local historic district designation compared to the 20th century.

6. Discussions

The shared concern for the conservation of urban neighborhoods between planners and preservationists (Birch & Roby, 1984) has contributed to the creation of thousands of local historic districts in U.S. cities, particularly over the last 50 years. The recent call for social justice within both fields raises the question of whether the use of local historic district designation is equitable and inclusive. While a large body of preservation literature concentrates on the impacts of local historic districts on neighborhood characteristics following their designation, little attention has been paid to this relationship at the time of designation. To fill this research gap, this study investigated whether local historic districts have been designated more or less frequently in spatially segregated neighborhoods across 32 large American cities since the 1970s. We used the HOLC neighborhood evaluation maps as a proxy of historical redlining and measured the Getis-Ord local *Gi** statistic to identify the recent patterns of spatial segregation. Our analysis using the *Gi** statistic showed significant associations with local historic district designations, whereas the results from the HOLC models remained largely inconclusive.

The results from our statistical models using the Gi* statistic indicate that local historic districts were more likely to be designated in racially and economically segregated neighborhoods. In other words, local historic districts were more prevalent in neighborhoods where African-Americans, Hispanics, and/or low-income populations were concentrated in each city at the time of district designation. Our findings contrast with existing critiques that argue the underrepresentation of marginalized neighborhoods in official historic listings (Gao & Dublin, 2020; Hayden, 1988; T. Lee, 2012). These results suggest that local historic districts have been utilized to preserve historic built environments in segregated neighborhoods. However, the extent to which the histories of populations that have encountered disadvantages are adequately represented in the designation of local historic districts remains unclear because our analysis only examines the relationship with district designation status. We are unable to demonstrate if the narratives and cultural heritage of underrepresented communities were thoroughly evaluated and fully recognized in the designation process of local historic districts.

A larger prevalence of local historic district designations in segregated neighborhoods provides evidence regarding the role of local historic districts as a neighborhood development strategy in late 20th-century American cities. This finding suggests that local historic districts were used as an alternative approach to demolition-based urban redevelopment in the 1970s (Birch & Roby, 1984; Hamer, 1998). It echoes several case studies that demonstrate the creation of local historic districts in economically distressed neighborhoods and racial and ethnic communities where urban renewal projects were planned (Hodder, 1996; S.P. Lee, 2001; Ryberg, 2011). In addition, the greater prevalence of local historic districts in economically segregated neighborhoods implies that locally designated historic districts were widely employed for community development and neighborhood revitalization, encouraging historic rehabilitation projects and promoting heritage tourism (Listokin et al., 1998; Ryberg-Webster & Kinahan, 2014). On the other hand, underserved populations in these local historic districts are more likely to be exposed to the threat of gentrification and displacement (Choi, 2025; Kinahan & Ruther, 2020; McCabe & Ellen, 2016; Zahirovic-Herbert & Chatterjee, 2012).

Although we find significant relationships between the designations of local historic districts and the local Getis-Ord *Gi** statistic from the 1970s to the 1990s, some of the results for the 2000s and 2010s within the *Gi** statistic analysis are inconclusive or reversed. This shift in the 21st century could be attributed to the increasing number of local historic district designations in low-segregation neighborhoods, while the number in high-segregation neighborhoods is stable or slightly reduced. Our findings suggest that the recently created local historic districts in the low-segregation neighborhoods could function as a mechanism for exclusionary zoning or restrictive covenants. Additional legal and economic burdens followed by local historic districts often lead to the exclusion and displacement of low-income housing and people of color (Fein, 1985). In addition, the designation of local historic districts would offer a tool to protect a single-family neighborhood from upzoning efforts to increase the supply of affordable housing. These findings suggest the need for additional case studies to explore potential conflicts between district designations and upzoning policies.

Contrary to the results from the *Gi** statistic analysis, the relationship between the creation of local historic districts and the HOLC grades appears to be mostly insignificant. These results did not reveal statistically significant differences in local historic district designations between historically redlined (grade D) and non-redlined (grade A, B, and C) neighborhoods. The insignificant results from the HOLC models align with a recent critique questioning the overreliance on the HOLC maps to understand spatial inequalities today (Markley, 2024). The HOLC grades are not simply static rankings and homogenous groups, but they reflect the spatial and temporal dynamics of neighborhoods and have considerable internal variability within the same grades. Moreover, the HOLC maps are less significant than the Federal Housing Administration mortgage risk maps in capturing the lingering effects of redlining on the patterns of residential segregation and neighborhood socioeconomic and housing characteristics (Fishback et al., 2021; Xu, 2022). In our study, the use of the local Getis-Ord *Gi** statistic of residential segregation measured by tract-level demographic and socioeconomic characteristics supplements the methodological limitations of the HOLC maps reflecting neighborhood conditions that occurred after the creation of the HOLC maps. Particularly in the grade D neighborhoods since the 1970s, the proportion of low-segregation neighborhoods has increased, while the ratio of high-segregation neighborhoods has decreased. Therefore, our findings suggest the complexity of using the HOLC maps to examine the enduring impacts of historical redlining on socioeconomic characteristics in contemporary urban neighborhoods.

Although our research protocol using two different datasets, HOLC maps and the Getis-Ord *Gi** statistic, helps understand the spatial patterns of local historic district designations in relation to historical redlining and residential segregation, there are some technical limitations in the scope and methods of this study. First, our statistical models do not fully account for the historic significance of neighborhoods typically evaluated in the review process to designate local historic districts. Despite the usefulness of data about the historic significance in modeling the factors that drive the designation of historic neighborhoods (Noonan & Krupka, 2010), we only included the ratio of old buildings due to the lack of data and difficulties in normalizing it. Only some of the cities have their own inventory of historic resources based on citywide comprehensive surveys. Even if all study cities surveyed historic resources, their results could be different in criteria, depth, scale, and period. Second, although the NCDB allows us to compare changing patterns of local historic district designations for the last five decades, it does not provide block group level census data or smaller spatial units for a fine-grained analysis (McCabe & Ellen, 2016). Thus, we exclude some local historic districts from our samples since census tracts are too large to represent their socioeconomic and housing characteristics. Lastly, further research on the spatial patterns of historic district designations could include additional large American cities with different contexts and histories of historic preservation. Several large cities were not included in our study because they were not assessed by HOLC or had relatively small populations at the time of the HOLC map creation.[7] However, expanding the study areas beyond the consideration of the HOLC maps would provide a more comprehensive understanding and valuable insights into how recent segregation patterns are associated with the designation of historic districts in a broader range of urban settings.

7. Conclusion

While thousands of local historic districts have been designated in American cities to protect neighborhood characteristics and manage their transformations over the last 50 years, a question is posed whether neighborhoods with specific population groups are more or less represented in the designations. Our study particularly focuses on the relationship between the designation of local historic districts and the persistent landscape of spatial segregation historically created and reinforced by institutional forces. While we employed both the HOLC maps and Getis-Ord *Gi** statistic to capture historical and recent spatial patterns of residential segregation, previous research has not comprehensively explored how preservation policies are associated with enduring socio-spatial inequalities in urban areas. This approach has broader applications for evaluating equitable and inclusive preservation practices not only in American metropolitan areas but also in other cities around the world with similar challenges in their socio-spatial dynamics. Our analysis revealed that segregated neighborhoods were more

[7] The large cities that were not assessed by HOLC include Washington, D.C., Charlotte, NC, Las Vegas, NV, and Cincinnati, OH, whereas the less populous cities at that time include Phoenix, AZ, Miami, FL, Tampa, FL, and Nashville, TN.

likely to establish local historic districts in the late 20th century, but this pattern started to change in the early 21st century. Our findings on the patterns in the 20th century challenge the conventional view on the underrepresentation of marginalized communities in historic designations. The changing patterns in the 21st century suggest the need for a broader exploration of equitable outcomes in terms of socio-spatial justice, such as exclusiveness, affordability, and density, in the recently growing discourse on the equitable preservation framework.

The findings from our analysis present the challenges for preservation planners and local landmark preservation boards to reconsider historic designation procedures through the equity preservation framework. In order to advance more equitable and inclusive preservation policies, they can pay more attention to whether the creation of local historic districts mitigates or reinforces the persistent landscape of spatial segregation in the designation process. Also, it is necessary to ensure equitable representation of historically marginalized communities and their histories and engage members from segregated neighborhoods in the preservation decision-making process. In addition, the growing number of local historic districts in low-segregation neighborhoods raises concerns about the possibility that these districts might serve as tools of exclusionary zoning, suggesting the need for cooperation with housing departments and agencies to consider their influence on the overall housing supply and affordability. These policy implications are also pertinent to international efforts for preservation planning and heritage management to achieve equity goals of sustainable development through promoting diversity and inclusion in historic designation and listing.

CRediT authorship contribution statement

Sunho Choi: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rebecca J. Walter:** Writing – review & editing, Supervision, Methodology. **Manish Chalana:** Writing – review & editing, Supervision.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

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Data availability

The authors do not have permission to share data.

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